Chanattan Sok

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École Normale Supérieure de Rennes, Bruz, France & Sorbonne Université – Paris, France

chanattan.sok@ens-rennes.fr | chanattan.github.io | linkedin.com/in/chanattan-sok | Born on the 6th of August 2002

Research Interest

I am interested in decision-making (aid) in general (sequential, collective), game-theory & mechanism design, LLMs (alignment), and more recently, Agentic AI. I work in interdisciplinary settings, including economics, social sciences with computer science for enhanced (decision) and innovative solutions.

Education

Master's Degree in Computer Science – AI2D (Artificial Intelligence, Algorithms, Interaction, Decision Making) - Sorbonne University, Paris, France — Sept 2025 - May 2026
Decision-making (sequential, collective, multi-criteria), RL & RLHF, mechanism design, game theory, and social choice.

ENS Diploma in Computer Science - École normale supérieure de Rennes, France — Aug 2022 - Aug 2026

B.Sc. in Fundamentals of Computer Science - École normale supérieure de Rennes — Aug 2022 – Aug 2023

2nd year of B.Sc. in Computer Science - Sorbonne University, Paris, France — Sept 2021 - May 2022
- Ranked 5th/172 in computer science and mathematics.

Preparatory classes to Grandes Écoles - Lycée Albert Schweitzer, Raincy, France — Sept 2020 - May 2021

Research Positions

Research Assistant, LIP6 (Sorbonne University, Paris) — March – Aug 2026
On the study of fairness and strategies in collective multi-criteria decision-making settings.
Supervised by Pr. Aurélie Beynier, Pr. Fanny Pascual, Assoc. Pr. Nawal Benabbou.
- Study and proof of convergence to equilibria in fair multi-criteria allocation of indivisible items, e.g., kidney attribution.
- Empirical experiments with artificial and real preferences on other decision-making settings, e.g., auction as a processus.

Research Assistant, LAMSADE (Dauphine–PSL, Paris) — May–Aug 2025
Fairness in railway networks: graph theory and social choice. Supervised by J. Lang, A. Novaro, F. Sikora. GitHub
- Developed a Python software to compute custom fairness indices on editable railway networks (article in writing).
- Proposed and evaluated fairness metrics on GIS-based real railway networks (e.g., France vs Germany).

Research Fellow, City Science Lab (HCU Hamburg - MIT Media Lab & IRD) — Feb–May 2025
Urban digital twins and ABM for climate policy in Hamburg (2040s). Supervised by A. Grignard, R. Herzog. GitHub
- GAMA-based urban model using GIS data for policy support; collab. with Hamburg Climate Authority.
- Presented at Digital Urban Twins 2025, and journal article in acceptation at the Journal of Urban Technology.

Research Assistant, CHROMA (Inria & INSA Lyon) — Sept–Dec 2024
Resolving pedestrian spatial conflicts with ABM & social choice. Paper at JFSMA 2025 with O. Simonin. GitHub
- Designed the theoretical and applied *CNM* model (negotiation-based pathfinding) outperforming SFM and CBS-MAPF.
- CNM shows up to 10 times faster-computed solutions than CBS-MAPF, and as efficient, with human-like behavior.

Research Assistant, Programming Languages Group (Aarhus Univ., Denmark) — May–Aug 2024
Java type inference and reduction in Flix. Supervised by M. Madsen.
- Proposed novel Java type representations; improved Flix programming language's modularity.
- Implemented Java interoperability in Flix compiler, made in Scala 2.

Research Assistant, SHAMAN (IRISA, Univ. Rennes & Thales) — Sept 2023–May 2024
WebAssembly for data management in serverless environments. With L. d'Orazio, R. Tekin, D. Tombroff GitHub
- First-author paper at SSDBM 2024 (<30% acceptance).
- Developed *Blossom*, a platform to benchmark & test out-of-the-web Wasm vs native Rust join operations.

Student Research Fellow, ENS Rennes & Anticor — Nov 2023–Now
AI-Law: LLMs (RAG, Mixture of Experts, deliberation), data (SQL, Pandas), UX/UI. With J. Perez. Press article
- *Project 1:* Designed assistant tool for lawyers for fraud detection in public procurement with LLM as a classifier.
- *Project 2:* RAG and post-training on law text corpus. Mixture of Experts with agent-as-judge and deliberation.

Research Assistant, MAS Team (LIP6, Sorbonne Univ., CNRS) — May–July 2022

ABM modeling of international (climate) negotiations. Supervised by N. Maudet, A. Beynier. GitLab

- Re-implemented and extended significantly Earnest's negotiation model (Putnam's two-level game) using Mesa in Python. Research nuanced counter-intuitions on multi-agent deliberation (the more people, the harder the debate).
- Joined as a participant in the NEGOCLIM project with CES economists (e.g., Zuber, Rusinowska).

Projects

NEGOCLIM project (summer 2023) Website
The project brings together researchers from economy, CS, humanities to study climate negotiations. It is funded by the CNRS. I participated during a research internship by proposing a generic negotiation model between agents.

Winner – Palm'Hackathon ISTIC, Univ. Rennes (2023) GitHub
Built a game in team in Rust on "investigation" theme in under 24h. Ranked 1st among 8 teams.

Freelance developer (2015-2021) — *over 25 repositories including hackathons, ABM models, and Java/Python/Rust/C++ projects* GitHub
For each of my work, I try to code a tool or software to automate protocols, and ease work continuation with visuals. Java Swing/FX automation tools: isometric map editor, scrapers, data pipelines, visualization & editing, e.g., graphs. Crypto-oriented blockchain project made in C, 2/2.5D games. *Thoth:* AI-assistant in simulated trading (ARIMA model).

Publications (excluding in processing)

Resolution and simulation of spatial conflicts between pedestrians based on social choice theory (HAL science – page 128) May 2025
Chanattan Sok, Olivier Simonin. PFIA, JFSMA 2025, Dijon, France (ECAI 2025 version, weakly rejected)

WebAssembly Serverless Join: A Study of its Application July 2024
Chanattan Sok, L. d'Orazio, R. Tekin, D. Tombroff. SSDBM 2024, Rennes
10.1145/3676288.3676305

Teaching

***"Colleur"* in preparatory classes MP2I/MPI in CS, EISAM – Cameroun** Feb 2026–Now

Tutor founder, Univ. Rennes – ISTIC Sept 2023–Mar 2024
Founded CS/maths tutoring program for bachelor students. Led and organized sessions.

Guest lecturer, CalPoly (California) Jan 2024
Lecture on ABM and negotiation models in MAS course. Invited by Pr. F. Kurfess.

CPES science tutor (maths/CS), Lycée Châteaubriand, Rennes Sept 2023–Mar 2024
Tutoring in selective preparation program (CPES).

Tutor in general science (Cordées de la Réussite), Rennes Sept 2023–Mar 2024
Mentored high school students from underrepresented backgrounds.

Teaching assistant, Collège Évariste Galois, Paris Sept 2021–Mar 2022
Taught math/science classes and assisted with classroom management. Selective program.

Private CS/Maths tutor (freelance) 2015–2024
Taught Python, Java OOP, C/C++, algorithms to students from middle school to university.

Additional Experiences and Activities

Hi! PARIS Summer School, École Polytechnique 2025
Tutorials on Reinforcement Learning, Recommender systems.

Co-president, Oenology Club, ENS Rennes Sept 2023–Mar 2024
Led wine-tasting events, presentations, and budget management.

Tutor Founder, Tutorat ISTIC, Univ. Rennes Sept 2023–Mar 2024
Organized CS/Maths tutoring sessions for B.Sc. students and liaised with administration.

Finalist, ENS Rennes Eloquence Contest. 2023

Treasurer, RJMI Rennes 2023 Oct 2022–Mar 2023
Led treasury management in a major STEM outreach event for high school girls, promoting CS and Maths excellence to advance equal opportunities in higher studies.

Languages

Tech: Java 7/8, Python, Scala 2, C, C++, SQL, OCaml, Rust, Coq, GAMS
Non-tech: French (native), English (proficient), German (notions)